Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		**A** $FeCl_3$ (1) **B** NaCl (1) **C** Br_2 (1) award (1) for correct **names** of all three substances		3		3		
		(ii)		fume cupboard – because chlorine / bromine / halogen fumes are toxic **three** parts needed neutral: reference to goggles or general laboratory safety or fumes in general	1			1		1
		(iii)		no reaction / no change / no observable change (1) iodine is less reactive than bromine (1) and is therefore unable to displace the bromide / bromine (from the solution/sodium bromide) (1) neutral: iodine less reactive than chlorine gains no credit	2	1		3		3

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
	(b)			ClO_2 (3) if formula incorrect award credit for correct steps $Cl = \frac{0.71}{35.5}/0.02$ **and** $O = \frac{0.64}{16}/0.04$ (1) conversion of 0.02 and 0.04 to a 1:2 ratio (1) ecf possible award (1) max if A_r divided by mass leading to Cl_2O		3		3	3	
	(c)	(i)		$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$ correct formulae for both ions and product (1) state symbols for both ions and product (1) state symbols only credited if ions and product correct	1	1		2		
		(ii)		$2AgNO_3 + CaI_2 \rightarrow 2AgI + Ca(NO_3)_2$ correct reactants (1) correct products (1) balancing (1) balancing mark can only be awarded if both the reactants and products are correct		3		3	1	
				Question 5 total	**4**	**11**	**0**	**15**	**4**	**4**